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Inventor(s)	Langer; Tamir et al.

POE PSE MPS support for PSE voltage transients

Abstract

An apparatus includes a power-over-Ethernet (POE) interface to be connected to a powered device (PD) over an Ethernet cable and a control circuit. The control circuit is to measure a voltage provided by the apparatus through the Ethernet cable, determine that the voltage has dropped by at least a given voltage change, based on a determination that the voltage has dropped by at least the given voltage change, determine whether or not a predetermined quantity of Maintain Power Signature (MPS) signals have been missed within a given time frame, and, based on a determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determine that the PD is still connected to the apparatus.

Inventors:	Langer; Tamir (Tel Aviv, IL), Peker; Arkadiy (Glen Cove, NY)
Applicant:	Microchip Technology Incorporated (Chandler, AZ)
Family ID:	1000008782050
Assignee:	Microchip Technology Incorporated (Chandler, AZ)
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2006/0181817	12/2005	Ohana	361/18	G06F 1/26
2008/0040625	12/2007	Vorenkamp	713/340	H04L 25/02
2015/0137618	12/2014	Vaajala	307/116	H02H 3/0935
2015/0372826	12/2014	Blaut	307/1	H04L 12/10
2016/0095175	12/2015	Picard et al.	315/294	N/A
2017/0012787	12/2016	Horváth et al.	307/1	N/A
2018/0176026	12/2017	Yseboodt et al.	N/A	N/A
2018/0227132	12/2017	Picard	N/A	H05B 47/187
2021/0194708	12/2020	Wendt	N/A	G05F 1/70
2023/0004202	12/2022	Yi	N/A	G06F 1/266

OTHER PUBLICATIONS

International Search Report and Written Opinion, Application No. PCT/US2023/028931, 13 pages, Nov. 3, 2023. cited by applicant

Primary Examiner: Bae; Ji H

Attorney, Agent or Firm: SLAYDEN GRUBERT BEARD PLLC

Background/Summary

PRIORITY (1) This application claims priority to U.S. Provisional Patent Application No. 63/392,950 filed Jul. 28, 2022, the contents of which are hereby incorporated in their entirety.

FIELD OF THE INVENTION

(1) The present disclosure relates to power over Ethernet (POE) and, more particularly, to POE power sourcing equipment (PSE) Maintain Power Signature (MPS) support for PSE voltage transients.

BACKGROUND

(2) POE may include passing electric power along with data on twisted-pair Ethernet cabling. This may allow a cable to provide both data connection and electric power from a source (referred to as PSE) to devices (referred to as powered devices (PD)). PD devices may include, for example,

wireless access points (WAPs), Internet Protocol (IP) cameras, and voice over Internet Protocol (VoIP) phones.

(3) When a PSE is supplying power to a PD, the PSE may monitor the current drawn by the PD in order to make sure that the PD is still connected. The minimum current that the PD has to draw to avoid being considered to be disconnected from the PSE is called a maintain power signature (MPS) signal. A spike in current or an instance in current drawn by the PD and as observed by the PSE may be considered an MPS signal. This behavior may be defined by, for example, the IEEE 802.3 bt standard, which further specifies amplitude and duration of MPS current that is drawn by PDs and observed by PSEs. For example, the PSE may remove power from the POE cable connecting the PSE to the PD when the PSE has not seen an MPS signal for at least 400 milliseconds, so that cables disconnected from the PD do not remain powered.

(4) The MPS signal may be defined as a current pulse received at the PSE from the PD that draws a certain amount of current for a minimum amount of time, with a certain duty-cycle. The amplitude, minimum pulse-width, and duty cycle may be defined in various different POE standards or protocols, and may be a function of the assigned class, a capability of the PSE to support short (low duty-cycle) or long (higher duty-cycle) MPS, and system characteristics (as cable resistance, DC/DC input capacitance, etc.).

(5) Inventors of examples of the present disclosure have discovered that a PD might not issue MPS pulses in order to remain powered. Such devices might operate with relatively low power consumption compared to other PDs. Such devices may internally draw or absorb MPS pulses to remain powered. PD devices may have a current sink or resistor that the PD can control so as to appear to remain connected to the PSE. When such a circuit of the PD is activated but with low load, the resulting current may be initially sourced mostly from a bulk capacitor of the PD. Inventors of examples of the present disclosure have discovered that if voltage provided by the PSE to the PD, known as PSE voltage, rapidly decreases during MPS PSE-even though the PD, a diode bridge will be reverse biased and all current will be taken from the PD capacitor, and this may continue until PSE and PD voltage become approximately equal. This can happen during PSE power supply voltage transients, and in this case the PSE may determine that there is no MPS current and the PSE may shut down the PD, even though PD is supplying MPS current.

(6) Examples of the present disclosure may address one or more of these issues. Examples of the present disclosure may address false POE PSE port shutdowns in case of PSE voltage transients when PSE operate in MPS mode.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates an apparatus for POE, according to examples of the present disclosure.

(2) FIG. 2 is another illustration an of apparatus for POE, according to examples of the present disclosure.

(3) FIG. 3 is another illustration of an apparatus for POE, according to examples of the present disclosure.

(4) FIG. 4 is an illustration of a PD, according to examples of the present disclosure.

(5) FIG. 5 is a timing diagram of operation of a PD, according to examples of the present disclosure.

(6) FIG. 6 is an illustration of example values from an experiment for a given cable, PD, and apparatus **100**, according to examples of the present disclosure.

(7) FIG. 7 is an illustration of how many allowed missed MPS pulses may be allowed versus a VPSE voltage drop before determining that a PD should no longer be powered, according to examples of the present disclosure.

(8) FIG. 8 is an illustration of an example method for POE MPS support for PSE voltage transients, according to examples of the present disclosure.

(9) FIG. 9 is an illustration of another example method for POE MPS support for PSE voltage transients, according to examples of the present disclosure.

DETAILED DESCRIPTION

(10) FIG. 1 illustrates an apparatus **100** for POE, according to examples of the present disclosure.

(11) Apparatus **100** may include interface **102** configured to provide communication from apparatus **100** to any suitable electronic device, such as PD **106**. Interface **102** may be implemented in any suitable manner, such as by analog circuitry, digital circuitry, control logic, instructions in a memory for execution by a processor, digital logic circuits programmed through hardware description language, application specific integrated circuits (ASIC), field programmable gate arrays (FPGA), programmable logic devices (PLD), or any suitable combination thereof, whether in a unitary device or spread over several devices. Interface **102** may be configured to provide communication in any suitable format, such as Ethernet. PD **106** may be implemented as any suitable electronic device that receives its power at least in part through POE from apparatus **100**. Apparatus **100** may be connected to PD **106** over any suitable connection, such as an Ethernet cable **104**.

(12) Apparatus **100** may include a control circuit **108**. Control circuit **108** may be implemented in any suitable manner, such as by analog circuitry, digital circuitry, control logic, instructions in a memory for execution by a processor, digital logic circuits programmed through hardware description language, ASIC, FPGA, PLD, or any suitable combination thereof, whether in a unitary device or spread over several devices.

(13) Control circuit **108** may be configured to measure a voltage provided by apparatus **100** to PD **106** and returned from PD **106**. The voltage provided by apparatus **100** to PD **106** may be given as VPSE. A current may also be provided by apparatus **100** to PD **106**, and may be given as IPSE. A current may be returned from PD **106** and may be given as IPSE. The voltage provided by apparatus **100** may be provided through interface **102** and Ethernet cable **104** to PD **106**. Similarly, the current returned from PD **106** to apparatus **100** may be provided through Ethernet cable **104**.

(14) Control circuit **108** may be configured to measure VPSE and IPSE through any suitable mechanism. Analog to digital converters (ADCs) may sample VPSE. IPSE may be measured by a current sense resistor within apparatus **100**.

(15) Control circuit **108** may be configured to measure VPSE and determine whether VPSE has dropped by at least a given voltage change, given by VTHRESHOLD. The voltage change may be given by VPSE.sub.1-VPSE.sub.2. The measurement may be made periodically or on-demand, such as every millisecond, though some sampling may occur faster or may occur every 1-5 milliseconds. The voltage change may be a voltage drop. The first and second measurements of VPSE may be made at any suitable moments in time. The first and second measurements of VPSE may be consecutive or may be nonconsecutive. Control circuit **108** may be configured to, based upon a determination that the given voltage change has exceeded the threshold, determine whether or not a predetermined quantity of MPS signals have been missed within a given time frame. The predetermined quantity may be given as MPS-THRESHOLD. The predetermined quantity of MPS signals may be greater than zero. Control circuit **108** may be configured to, based on a determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determine that PD **106** is still connected to apparatus **100**.

(16) The given time frame may be of any suitable length. The voltage drop threshold may be of any suitable value. The predetermined quantity of MPS signals that have been missed within the given time frame may be of any suitable value. Moreover, the predetermined quantity of MPS signals may be based upon the value of the voltage drop, and may vary according to the value of the voltage drop.

(17) Any suitable manner or mechanism may be used to determine a quantity of missed MPS

signals. For example, a counter (not shown) may be used, and at each point in time when an MPS signal was expected but was not received, the counter may be incremented by control circuit **108**. If the predetermined quantity is reached, then control circuit **108** may make the appropriate determination. In another example, the time to wait before determining whether or not the predetermined quantity of MPS signals has been missed may be equal to the predetermined quantity times the total cycle time of a given MPS signal. In such an example, control circuit **108** may wait for such a time to wait to determine whether an MPS signal has yet been received, without explicitly counting the missed MPS signals, as such a time to wait may implicitly reflect such a number of missed MPS signals.

(18) Thus, control circuit **108** may cause apparatus **100** to continue to power PD **106** even if an MPS signal is missed, at least for a given count of missed MPS signals greater than zero. Moreover, control circuit **108** may measure VPSE and determine a voltage drop and, based on the value of the voltage drop, determine a quantity of MPS signals that are allowed to be missed at apparatus **100** before powering down PD **106**. Thus, apparatus **100** might not falsely turn off PD **106** in case of a voltage drop.

(19) FIG. **2** is another illustration of apparatus **100** for POE, according to examples of the present disclosure.

(20) Control circuit **108** may be configured to determine VPSE and VPSE. Control circuit **108** may be configured to determine whether VPSE has dropped by at least a given voltage change, given by VTHRESHOLD. The voltage drop may be given by VPSE.sub.1-VPSE.sub.2. Control circuit **108** may be configured to, based upon a determination that the given voltage change has exceeded the threshold, determine whether or not a predetermined quantity of MPS signals have been missed within a given time frame. The predetermined quantity may be given as MPS-THRESHOLD. The predetermined quantity of MPS signals may be greater than zero. Control circuit **108** may be configured to, based on a determination that the predetermined quantity of MPS signals has been missed within the given time frame, determine that PD **106** is disconnected from apparatus **100**.

(21) The given time frame may be of any suitable length. The voltage drop threshold may be of any suitable value. The predetermined quantity of MPS signals that have been missed within the given time frame may be of any suitable value. Moreover, the predetermined quantity of MPS signals may be based upon the value of the voltage drop, and may vary according to the value of the voltage drop.

(22) Any suitable manner or mechanism may be used to determine a quantity of missed MPS signals. For example, a counter (not shown) may be used, and at each point in time when an MPS signal was expected but was not received, the counter may be incremented by control circuit **108**. If the predetermined quantity is reached, then control circuit **108** may make the appropriate determination. In another example, the time to wait before determining whether or not the predetermined quantity of MPS signals has been missed may be equal to the predetermined quantity times the total cycle time of a given MPS signal. In such an example, control circuit **108** may wait for such a time to wait to determine whether an MPS signal has yet been received, without explicitly counting the missed MPS signals, as such a time to wait may implicitly reflect such a number of missed MPS signals.

(23) FIG. **3** is another illustration of apparatus **100** for POE, according to examples of the present disclosure. Apparatus **100** is illustrated with a counter **110** configured to count missed MPS signals. Counter **110** may be implemented in any suitable manner, and may be implemented within or without control circuit **108**.

(24) Control circuit **108** may be configured to determine VPSE and VPSE. Control circuit **108** may be configured to measure VPSE.sub.1 and VPSE.sub.2 and determine whether VPSE has dropped by at least a given voltage change, given by VTHRESHOLD. The voltage change may be given by VPSE.sub.1-VPSE.sub.2. Control circuit **108** may be configured to, based upon a determination that the given voltage change has exceeded the threshold, set counter **110** to count missing MPS

signals. An MPS signal may be determined to have been missed if a given time cycle for an expected MPS signal from PD **106** has expired without receiving an MPS signal. After setting counter **110** to count missing MPS signals, control circuit **108** may be configured to determine whether an additional MPS signal has been received. If such an additional MPS signal is received, control circuit **108** may be configured to determine that PD **106** is still connected to apparatus **100**. If an additional MPS signal has not been received, control circuit **108** may be configured to determine whether counter **110** has reached the predetermined quantity of MPS signals that may be missed within a given time frame, given as MPSTHRESHOLD. If counter **110** has reached MPSTHRESHOLD, control circuit **108** may be configured to determine that PD **106** is disconnected from apparatus **100**. MPSTHRESHOLD may be greater than zero.

(25) FIG. **4** is an illustration of PD **106**, according to examples of the present disclosure.

(26) PD **106** may include a front-end connector **402**. Connector **402** may be configured to receive VPSE from apparatus **100** and provide IPSE to apparatus **100**. Such portions may include transformer pairs to which the rest of PD **106** may be tapped. Connector **402** may include two ports. In the present disclosure, only the first port is discussed, for the purposes of simplicity.

(27) PD **106** may include diode structures **404**, implemented with any suitable number, kind, and arrangement of diodes. Diode structures **404** may act as diode bridges.

(28) PD **106** may include a transistor **414**. Transistor **414** may be implemented by any suitable transistor implementation. Transistor **414** may be configured to isolate PD **106** from PD load **106** during PD detection and classification.

(29) PD **106** may include a PD controller **406**. PD controller **406** may be implemented in any suitable manner, such as by analog circuitry, digital circuitry, control logic, instructions in a memory for execution by a processor, digital logic circuits programmed through hardware description language, ASIC, FPGA, PLD, or any suitable combination thereof, whether in a unitary device or spread over several devices. PD controller **406** may also include resistor **410**, transistor **408**, and transistor **414**. PD controller **406** may be configured to cause an MPS signal to be generated periodically. The MPS signal may be provided to apparatus **100** (when connected) directly or indirectly. PD controller **106** may cause a transistor **408** (implemented in any suitable manner) to turn ON during the MPS signal on-time. This may cause a voltage supply, such as VPSE, to be provided through a resistor **410**. This may cause current to flow out to apparatus **100** as IPSE, which may flow through resistor **410**.

(30) PD **106** may include a bulk capacitor **412** in parallel with a load of PD **106**, given by PD load **416**, and in parallel with diode structures **404**.

(31) In other implementations, a PD may need to be implemented with a limit or reduced size for a bulk capacitor where PSE voltage transients may occur. For example, a TEXAS INSTRUMENTS PD IC TPS2373 may specify a maximum capacitance of 300 uF for PSE voltage step-downs below 0.4 V, or a maximum capacitance of 120 uF for larger voltage step-downs such as 57V-50V. This is despite the fact that the POE specifications allow such capacitors to be up to 360 uF capacity. Thus, as a practical matter, in other implementations the size of a bulk capacitor is not as big as the size could be. By limiting capacitance values, the performance may be degraded and unwanted and unknown connection requirements may exist between a given PSE and PD. In contrast, examples of the present disclosure may remove dependencies between apparatus **100** (operated as a PSE) and PD **106** and allow PD **106** to use any values of capacitors, regardless of any PSE voltage transients.

(32) For various implementations of PD **106**, a minimum duty-cycle of MPS may be specified. Moreover, MPS pulse amplitude and width may be measured at apparatus **100** with assumptions of use of maximum cable length and worst-case capacitance. As a result, some required timings may reflect the time and amplitude at apparatus **100** of the IPSE to be observed.

(33) During voltage transients that affect the available supply of voltage in PD **106**, the diode bridges in diode portions **404** of PD **106** may become reverse-biased and IPSE might not be generated from VPSE within PD **106**, but instead by bulk capacitor **412**, which might otherwise

cause apparatus **100** to believe that there is no IPSE and to turn off power to PD **106**. PD **106** may include a current sink or resistor that it can control, in the form of resistor **410** and transistor **408**. When such a current sink or resistor is activated during the voltage transient, the resulting current may be initially mostly sourced from bulk capacitor **412** rather than from VPSE from apparatus **100**, because the diodes of diode portions **404** are reverse-biased. As the voltage of bulk capacitor **412** drops, more current may be drawn from apparatus **100** the PSE until finally the voltages from bulk capacitor **412** and VPSE have equalized and all current is drawn from apparatus **100**. When VPSE rapidly decreases when an MPS signal is expected, apparatus **100** might not see the associated IPSE spike because a diode bridge such as those in diode structure **404** may be reverse biased and all current may be taken from bulk capacitor **412**. Furthermore, this may continue until the voltages from bulk capacitor **412** and VPSE have equalized. This may describe a power supply voltage transient, and in this case the MPS signal would not be propagated to apparatus **100**. In other implementations, the PSE would consider that there is no MPS current and the PSE would shut down the PD even though the PD is still connected and supplying MPS current. This may be a transient voltage spike or sink wherein the PD does not draw as much voltage from the PSE as the PD is actually using and may thus appear, from the perspective of the PD, to be disconnected from the PD. This may cause the PSE to falsely disconnect the PD from power. However, apparatus **100** may avoid this problem by first detecting a voltage drop on VPSE, and then allow a certain number of missed MPS signals before determining to disconnect PD **106** from power.

(34) FIG. 5 is a timing diagram of operation of a PD such as PD **106**, according to examples of the present disclosure.

(35) Illustrated are graphs of VPSE (as internal to PD **106**), VPD (the voltage across PD **106**), output of PD controller **406** or attempted MPS pulses, the current through bulk capacitor **412**, the current IPSE out of PD **106** to apparatus **100**, PSE MPS detection at apparatus **100**, and an interpretation of whether MPS signals have been received within the designated time frame.

(36) As shown, VPSE within PD **106** may drop due to a voltage transient condition, or due to a change of the total load on the power supply of the PSE, VPSE. As VPSE drops, bulk capacitor **412** may not allow VPD to abruptly discharge, resulting in VPD being higher than VPSE and forcing diodes of diode portion **404** to become reverse biased. PD controller **406** may issue MPS signals in the form of pulses. The time between such pulses may be less than a given constant. The constant may be given according to a particular standard followed by apparatus **100** or PD **106**, and may be given as tMPDO. The current through bulk capacitor **412** may correspond to pulses generated by PD controller **406** output, but only while VPD is above VPSE less the forward voltage of diode portions **404**. Such a forward voltage may be typically equal to two times the voltage drop (VD) of each diode comprising the diode bridges of diode portions **404** (at least for passive diodes). The current drawn by the MPS generated in PD **106** may discharge bulk capacitor **412**, resulting in a decrease of VPD with every MPS pulse, corresponding to the MPS current amplitude and bulk capacitor **412** capacitance value. The current applied out to apparatus **100**, IPSE, may follow the pulses generated to apparatus **100** until VPD discharges to the quantity VPSE less the forward voltage of the diodes of diode portions **404**. As shown, the current applied out to apparatus **100** might not be performed after the fall of VPSE and until VPSE and VPD are equal to within VDB (the forward voltage of diode portions **404**) again.

(37) MPS detection on apparatus **100** might detect the first pulse, as the current applied out to apparatus **100** from PD **106** may match the amplitude or duration requirements. However, the next two pulses may be missed, as no current might be applied out to apparatus **100** from PD **106**. Two such time periods, given by tMPDO, may elapse without receipt of such current pulses at apparatus **100**. Thus, in these time frames, the MPS may be considered to be absent.

(38) Once VPSE and VPD are equal to within VDB, the diodes comprising of the diode bridge **414** will become forward biased again, and the MPS pulses provided by PD controller **406** may again cause current pulses to be issued from PD **106** to apparatus **100**. Such current pulses may be

received at apparatus **100** and detected, and the MPS may be considered to be present.
(39) Thus, as shown it is possible for MPS signals to be missed while PD **106** is still connected to apparatus **100**, and after some MPS signals are missed at apparatus **100**, the MPS signals may yet return.

(40) FIG. **6** is an illustration of example values from an experiment for a given cable **104**, PD **106**, and apparatus **100** about the relationship between VPSE drops and how many MPS signals might be missed before MPS signals resume from PD **106** and apparatus **100**, according to examples of the present disclosure. Such analysis might be performed for other instances of cable **104**, PD **106**, and apparatus **100**, and the values shown in FIG. **6** are provided for demonstration purposes.

(41) Each row shows a different possible voltage drop from an original value of VPSE being 56 V. The permutations of FIG. **6** illustrate behavior for additional rows of voltage drops of increasing magnitude, wherein each row increases the voltage drop by 0.2 V. The second and third columns illustrate the VPSE value and the associated VPSE voltage drop. The next five columns illustrate a duration of the next five MPS pulses that are received by apparatus **100**. The next five columns illustrate whether such respective pulses would be missed by apparatus **100**. The next column is a repeat of the third column, for reference. The final column tallies the missed pulses.

(42) As shown in row 1, when VPSE is 56 V, and does not drop, no pulses might be missed. Each pulse may be received at apparatus **100** with a duration of 18.7 ms, and thus may be detected and not missed. No pulses might be missed in row 1.

(43) Similarly, in row 2, when VPSE was 56 V and drops 0.2 V, no pulses might be missed. Even though the first pulse may be of a shortened duration, it may be sufficiently long so as to be detected.

(44) However, in row 3, when VPSE was 56 V and drops 0.4 V, a pulse might be missed. The duration of the first pulse may be shortened to 6.75 ms, which might be too short for detection.

(45) In row 13, when VPSE was 56 V and drops 2.4 V, a second pulse might be missed. The second pulse may be shortened to 5.32 ms, which might be too short for detection.

(46) In row 23, when VPSE was 56 V and drops 4.4 V, a third pulse might be missed. The second pulse may be shortened to 3.47 ms, which might be too short for detection. In contrast, in row 22 immediately above row 23, when VPSE dropped by 4.2 V, the third pulse may have been 8.95 ms, which may have been long enough for detection.

(47) Accordingly, from such an example, set, control circuit **108** may be configured or programmed to determine how many MPS signals may be missed before disconnecting PD **106**, and such a quantity of MPS signals may be based upon the voltage drop observed for VPSE.

(48) FIG. **7** is an illustration of how many allowed missed MPS pulses may be allowed versus a VPSE voltage drop before determining that a PD should no longer be powered, according to examples of the present disclosure. This is based upon the example set of FIG. **6**.

(49) As shown, for a voltage drop between 0 and 0.4 V, no missed MPS signals should be experienced. Thus, a missed MPS signal in an associated voltage drop should be handled in a normal way—that is, it should indicate to apparatus **100** that PD **106** has disconnected, and power should be removed from PD **106**.

(50) For a voltage drop between 0.4 V and 2.4 V, a single missed MPS signal can be experienced while PD **106** is still connected. Thus, a single missed MPS signal with an associated voltage drop between 0.4 V and 2.4 V should not result in a determination that PD **106** has disconnected, and only after two missed MPS signals are detected should it be interpreted at apparatus **100** that PD **106** has disconnected, and power should be removed from PD **106**.

(51) For a voltage drop between 2.4 V and 4.4 V, two missed MPS signals can be experienced while PD **106** is still connected. Thus, one or two missed MPS signals in an associated voltage drop between 2.4 V and 4.4 V should not result in a determination that PD **106** has disconnected, and only after three missed MPS signals are detected should it be interpreted at apparatus **100** that PD **106** has disconnected, and power should be removed from PD **106**.

(52) For a voltage drop above 4.4 V, three missed MPS signals can be experienced while PD **106** is still connected. Thus, one, two, or three missed MPS signals with an associated voltage drop above 4.4 V should not result in a determination that PD **106** has disconnected, and only after four missed MPS signals are detected should it be interpreted at apparatus **100** that PD **106** has disconnected, and power should be removed from PD **106**.

(53) FIG. **8** is an illustration of an example method **800** for POE MPS support for PSE voltage transients, according to examples of the present disclosure. Method **800** may be performed by any suitable mechanism or element, such as by apparatus **100** or control circuit **108** therein. Method **800** may begin at any suitable step, such as **805**. The steps of method **800** may be repeated, omitted, performed in parallel, performed recursively, or in any suitable order with respect to one another.

(54) At **805**, it may be determined whether a PD device, such as PD **106**, is connected to a PSE apparatus, such as apparatus **100**. If so, method **800** may proceed to **810**. Otherwise, method **800** may return to **805**.

(55) At **810**, a VPSE such as VPSE supplied by the PSE apparatus to the PD device may be measured and apparatus.

(56) At **815**, it may be determined whether the VPSE has dropped by at least a given amount. The amount may be, for example, a range between $((n-1)*\Delta V + \text{offset})$ and $((n)*\Delta V + \text{offset})$. The possible range of VPSE may be divided into m sections each of a width of ΔV . The denotation of the section by n is a section in which one or more MPS signals might be missed without disconnecting or powering down a PD device. For example, in the scheme identified in FIG. **7**, ΔV may be 2 V and offset may be 0.4 V. A drop between 0.4 V and 2.4 V may correspond to $n=2$. A drop between 2.4 V and 4.4 V may correspond to $n=3$. A drop above 4.4 V may correspond to $n=3$. An out-of-range measurement above the set range may default to the $n=3$ case. For a given voltage drop, a lookup table or other similar data structure or mechanism may be used to determine an allowed number of missed MPS signals for the given voltage drop.

(57) If so, method **800** may proceed to **850**. Otherwise, method **800** may proceed to **820**.

(58) At **820**, a typical MPS detection scheme may be employed. A timer may be started. At **830**, it may be determined whether an MPS pulse or signal has been received. If so, method **800** may proceed to **830**. Otherwise, method **800** may proceed to **840**. At **840**, it may be determined whether a designated time period for an expected MPS pulse or signal has been reached, given as tMPDO. If so, method **800** may proceed to **845**. Otherwise, method **800** may return to **830**.

(59) At **835**, it may be determined that the PD device is still connected. PSE operation and powering of the PD device may continue. Method **800** may return to **810**.

(60) At **845**, it may be determined that the PD device is not connected. PSE operation and powering of the PD device may be turned off. Method **800** may return to **805**.

(61) At **850**, a modified MPS detection scheme may be employed wherein one or more MPS signals may be missed without determining that the PD is disconnected. A timer may be set. A counter to count missed MPS signals may be initialized.

(62) At **855**, it may be determined whether an MPS signal or pulse has been received. If so, method **800** may proceed to **835**. Otherwise, method **800** may proceed to **860**.

(63) At **860**, it may be determined whether a period of time for the expected MPS signal has been exceeded since an initialization or last update of missed MPS signal count. This period of time may be tMPDO, or may be tMPDO plus the expected duration of time of an MPS signal. This period of time may be given as Tdelay. If so, method **800** may proceed to **865**. Otherwise, method **800** may proceed to **870**.

(64) At **865**, the count of missed MPS signals may be incremented.

(65) At **870**, it may be determined whether the number of missed signals is at n or more, or whether the total amount of time for n missed signals (given by $(n+1)*T\text{delay}$) has been exceeded. If so, method **800** may proceed to **845**. Otherwise, method **800** may return to **855**.

(66) Method **800** may be amended so that specific values or ranges of voltage drops, associated with specific allowable missed MPS signals, may be determined. For example, specific checks of ranges may be made instead of the general form described in **815**. The ranges might not be linearly spaced, and so specific range checks, associated with specific counts of MPS signals that might be safely missed, may be used instead.

(67) FIG. **9** is an illustration of another example method **900** for POE MPS support for PSE voltage transients, according to examples of the present disclosure. Method **900** may be a more specific example of method **800**. The example of method **900** may utilize the example data from FIG. **6**.

(68) Method **900** may be performed by any suitable mechanism or element, such as by apparatus **100** or control circuit **108** therein. Method **900** may begin at any suitable step, such as **805**. The steps of method **900** may be repeated, omitted, performed in parallel, performed recursively, or in any suitable order with respect to one another.

(69) At **905**, it may be determined whether a PD device, such as PD **106**, is connected to a PSE apparatus, such as apparatus **100**. If so, method **900** may proceed to **910**. Otherwise, method **900** may return to **905**.

(70) At **910**, a counter for missed MPS pulses or signals may be cleared or initialized.

(71) At **915**, a VPSE such as VPSE supplied by the PSE apparatus to the PD device may be measured. The measurement may be averaged. The measurement may be compared with previous measurements to determine whether VPSE has dropped by a given amount.

(72) At **920**, it may be determined whether the voltage drop is within a first range. For example, it may be determined whether the voltage drop is between 0.4 V and 2.4 V. If so, method **900** may proceed to **925**. Otherwise, method **900** may proceed to **930**.

(73) At **925**, a threshold for missed MPS signals or pulses, such as MPS-THRESHOLD, may be set to one. This may enable a modified MPS detection scheme to be employed wherein one or more MPS signals may be missed without determining that the PD is disconnected. The number of allowable missed MPS signals may be based upon the voltage drop and may be determined by, for example, looking up the value in a lookup table. Method **900** may proceed to **955**.

(74) At **930**, it may be determined whether the voltage drop is within a second range. For example, it may be determined whether the voltage drop is between 2.4 V and 4.4 V. If so, method **900** may proceed to **935**. Otherwise, method **900** may proceed to **940**.

(75) At **935**, a threshold for missed MPS signals or pulses, such as MPS-THRESHOLD, may be set to two. This may enable a modified MPS detection scheme to be employed wherein one or more MPS signals may be missed without determining that the PD is disconnected. The number of allowable missed MPS signals may be based upon the voltage drop and may be determined by, for example, looking up the value in a lookup table. Method **900** may proceed to **955**.

(76) At **940**, it may be determined whether the voltage drop is within a third range. For example, it may be determined whether the voltage drop is greater than 4.4 V. If so, method **900** may proceed to **945**. Otherwise, method **900** may proceed to **950** if there are no other ranges to consider.

(77) At **945**, a threshold for missed MPS signals or pulses, such as MPS-THRESHOLD, may be set to three. This may enable a modified MPS detection scheme to be employed wherein one or more MPS signals may be missed without determining that the PD is disconnected. The number of allowable missed MPS signals may be based upon the voltage drop and may be determined by, for example, looking up the value in a lookup table. Method **900** may proceed to **955**.

(78) At **950**, a typical MPS detection scheme may be employed by setting MPS-THRESHOLD to zero. Method **900** may proceed to **955**.

(79) At **955**, a timer may be started. The timer may be set to a maximum time to wait for an MPS pulse or signal.

(80) At **960**, it may be determined whether an MPS pulse or signal has been received. If so, method **900** may proceed to **990**. Otherwise, method **900** may proceed to **965**.

(81) At **965**, it may be determined whether a designated time period for an expected MPS pulse or

signal has been reached, given as tMPDO, or where there has been one missed MPS signal. If so, method **900** may proceed to **970**. Otherwise, method **900** may return to **960**.

(82) At **970**, the missed MPS pulse counter may be incremented by one.

(83) At **975**, it may be determined whether a certain quantity of MPS signals been missed (or time elapsed since an MPS signal was received, such a time equivalent to the certain quantity of MPS signals). The quantity of MPS signals may be given by MPS-THRESHOLD and may have been defined in **925**, **935**, **945**, or **950**. Thus it may be determined whether the counter of missed MPS pulses or signals (or the equivalent time thereof) is greater than or equal to MPS-THRESHOLD (or the equivalent time thereof). If so, method **900** may proceed to **985**. Otherwise, method **900** may proceed to **980**.

(84) At **980**, the MPS timer for timing the duration between MPS signals may be reset. Method **800** may return to **955**.

(85) At **985**, it may be determined that the PD device is not connected. PSE operation and powering of the PD device may be turned off. Method **900** may return to **905**.

(86) At **990**, it may be determined that the PD device is still connected. PSE operation and powering of the PD device may continue. Method **900** may return to **910**.

(87) Examples of the present disclosure may include an apparatus. The apparatus may include a POE interface to be connected to a PD over an Ethernet cable. The apparatus may include a control circuit. The control circuit may be communicatively coupled to the POE. The control circuit may be configured to measure a voltage provided by the apparatus through the Ethernet cable. The voltage may be averaged. The voltage may be measured in subsequent measurements. The control circuit may be configured to determine that the voltage has dropped by at least a given voltage change. The control circuit may be configured to, based on a determination that the voltage has dropped by at least the given voltage change, determine whether or not a predetermined quantity of MPS signals have been missed within a given time frame. The predetermined quantity of MPS signals may be greater than zero. The control circuit may be configured to, based on a determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determine that the PD is still connected to the apparatus.

(88) In combination with any of the above embodiments, the predetermined quantity of MPS signals may be based upon on an amount of the voltage drop.

(89) In combination with any of the above embodiments, the predetermined quantity of MPS signals may be a first MPS quantity when the given voltage change is a first voltage change, and the predetermined quantity of MPS signals may be a second MPS quantity when the given voltage change is a second voltage change, wherein the first MPS quantity is less than the second MPS quantity and the first voltage change is less than the second voltage change.

(90) In combination with any of the above embodiments, the predetermined quantity of MPS signals may be a third MPS quantity when the given voltage change is a third voltage change, wherein the third MPS quantity is greater than the second MPS quantity and the third voltage change is greater than the second voltage change.

(91) In combination with any of the above embodiments, the control circuit may be configured to, based on a determination that the predetermined quantity of MPS signals have not been missed within the given time frame, determine that the PD is still connected to the apparatus, wherein at least one MPS signal was missed within the given time frame.

(92) In combination with any of the above embodiments, the control circuit may be configured to, based on the determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determine that the given voltage change was a transient voltage drop of the apparatus.

(93) In combination with any of the above embodiments, the control circuit may be configured to, based on a determination that at least one MPS signal was missed within the given time frame and that the predetermined quantity of MPS signals has not been missed within the given time frame,

determine that the MPS signals were prevented from reaching the apparatus by reverse-biased diodes in the PD.

(94) In combination with any of the above embodiments, the given time frame may be a first time duration when the given voltage change is a first voltage change, the given time frame may be a second time duration when the given voltage change is a second voltage change, and the first time duration may be less than the second time duration and the first voltage change is less than the second voltage change.

(95) In combination with any of the above embodiments, the control circuit may be configured to, based on a determination that the predetermined quantity of MPS signals was missed within the given time frame, determine that the PD is has disconnected from the apparatus.

(96) In combination with any of the above embodiments, the control circuit may be configured to count missed MPS signals with a counter, incrementing the count upon a elapsed time for each MPS signal.

(97) In combination with any of the above embodiments, the control circuit may be too configured to count missed MPS signals according to a total time without an MPS signal.

(98) In combination with any of the above embodiments, the control circuit may be configured to, based on a determination that the predetermined quantity of consecutive MPS signals has not been missed, determine that the PD is still connected to the apparatus.

(99) Examples of the present disclosure may include an apparatus with any of the elements of any of the above embodiments. The apparatus may include a counter configured to count missing MPS signals. The apparatus may include a control circuit configured to determine that a voltage provided by the apparatus to a PD has dropped by at least a given voltage change. The control circuit may be configured to, based upon a determination that the voltage provided to the PD has dropped by at least a given voltage change, set the counter to begin counting missing MPS signals. The control circuit may be configured to determine whether an additional MPS signal has been received. The control circuit may be configured to, based upon a determination that an additional MPS signal has been received, determine that the PD is still connected to the apparatus. The control circuit may be configured to, based upon a determination that an additional MPS signal has not been received, determine whether the counter has reached a predetermined quantity of missing MPS signals. The control circuit may be configured to, based upon a determination that the counter has a reached the predetermined quantity of missing MPS signals, determine that the PD is disconnected from the apparatus.

(100) Although examples have been described above, other variations and examples may be made from this disclosure without departing from the spirit and scope of these examples.

Claims

1. An apparatus, comprising: a power-over-Ethernet (POE) interface to be connected to a powered device (PD) over an Ethernet cable; and a control circuit communicatively coupled to the POE interface, the control circuit configured to: measure a voltage provided by the apparatus through the Ethernet cable; determine that the voltage has dropped by at least a given voltage change; based on a determination that the voltage has dropped by at least the given voltage change, determine whether or not a predetermined quantity of Maintain Power Signature (MPS) signals have been missed within a given time frame, the predetermined quantity of MPS signals greater than zero; and based on a determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determine that the PD is still connected to the apparatus; wherein the predetermined quantity of MPS signals is based upon an amount of the given voltage change.
2. The apparatus of claim 1, wherein the control circuit is configured to, based on the determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determine that the given voltage change was a transient voltage drop of the apparatus.

3. The apparatus of claim 1, wherein the control circuit is configured to, based on a determination that the predetermined quantity of MPS signals was missed within the given time frame, determine that the PD is disconnected from the apparatus.
4. An apparatus, comprising: a power-over-Ethernet (POE) interface to be connected to a powered device (PD) over an Ethernet cable; and a control circuit communicatively coupled to the POE interface, the control circuit configured to: measure a voltage provided by the apparatus through the Ethernet cable; determine that the voltage has dropped by at least a given voltage change; based on a determination that the voltage has dropped by at least the given voltage change, determine whether or not a predetermined quantity of Maintain Power Signature (MPS) signals have been missed within a given time frame, the predetermined quantity of MPS signals greater than zero; and based on a determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determine that the PD is still connected to the apparatus; wherein: the predetermined quantity of MPS signals is a first MPS quantity when the given voltage change is a first voltage change; the predetermined quantity of MPS signals is a second MPS quantity when the given voltage change is a second voltage change; and the first MPS quantity is less than the second MPS quantity and the first voltage change is less than the second voltage change.
5. The apparatus of claim 4, wherein: the predetermined quantity of MPS signals is a third MPS quantity when the given voltage change is a third voltage change; and the third MPS quantity is greater than the second MPS quantity and the third voltage change is greater than the second voltage change.
6. An apparatus, comprising: a power-over-Ethernet (POE) interface to be connected to a powered device (PD) over an Ethernet cable; and a control circuit communicatively coupled to the POE interface, the control circuit configured to: measure a voltage provided by the apparatus through the Ethernet cable; determine that the voltage has dropped by at least a given voltage change; based on a determination that the voltage has dropped by at least the given voltage change, determine whether or not a predetermined quantity of Maintain Power Signature (MPS) signals have been missed within a given time frame, the predetermined quantity of MPS signals greater than zero; and based on a determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determine that the PD is still connected to the apparatus; wherein the control circuit is configured to, based on the determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determine that the PD is still connected to the apparatus, wherein at least one MPS signal was missed within the given time frame.
7. An apparatus, comprising: a power-over-Ethernet (POE) interface to be connected to a powered device (PD) over an Ethernet cable; and a control circuit communicatively coupled to the POE interface, the control circuit configured to: measure a voltage provided by the apparatus through the Ethernet cable; determine that the voltage has dropped by at least a given voltage change; based on a determination that the voltage has dropped by at least the given voltage change, determine whether or not a predetermined quantity of Maintain Power Signature (MPS) signals have been missed within a given time frame, the predetermined quantity of MPS signals greater than zero; and based on a determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determine that the PD is still connected to the apparatus; wherein the control circuit is configured to, based on a determination that at least one MPS signal was missed within the given time frame and that the predetermined quantity of MPS signals has not been missed within the given time frame, determine that the at least one MPS signal was prevented from reaching the apparatus by reverse-biased diodes in the PD.
8. An apparatus, comprising: a power-over-Ethernet (POE) interface to be connected to a powered device (PD) over an Ethernet cable; and a control circuit communicatively coupled to the POE interface, the control circuit configured to: measure a voltage provided by the apparatus through the Ethernet cable; determine that the voltage has dropped by at least a given voltage change; based on a determination that the voltage has dropped by at least the given voltage change, determine

whether or not a predetermined quantity of Maintain Power Signature (MPS) signals have been missed within a given time frame, the predetermined quantity of MPS signals greater than zero; and based on a determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determine that the PD is still connected to the apparatus; wherein: the given time frame is a first time duration when the given voltage change is a first voltage change; the given time frame is a second time duration when the given voltage change is a second voltage change; and the first time duration is less than the second time duration and the first voltage change is less than the second voltage change.

9. A method, comprising: connecting, through a power-over-Ethernet (POE) interface, to a powered device (PD) over an Ethernet cable; measuring a voltage provided through the Ethernet cable; determining that the voltage has dropped by at least a given voltage change; based on a determination that the voltage has dropped by at least the given voltage change, determining whether or not a predetermined quantity of Maintain Power Signature (MPS) signals have been missed within a given time frame, the predetermined quantity of MPS signals greater than zero; and based on a determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determining that the PD is still connected; wherein the predetermined quantity of MPS signals is based upon the given voltage change.

10. The method of claim 9, comprising, based on a determination that the predetermined quantity of MPS signals was missed within the given time frame, determining that the PD has disconnected.

11. A method, comprising: connecting, through a power-over-Ethernet (POE) interface, to a powered device (PD) over an Ethernet cable; measuring a voltage provided through the Ethernet cable; determining that the voltage has dropped by at least a given voltage change; based on a determination that the voltage has dropped by at least the given voltage change, determining whether or not a predetermined quantity of Maintain Power Signature (MPS) signals have been missed within a given time frame, the predetermined quantity of MPS signals greater than zero; and based on a determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determining that the PD is still connected; wherein: the predetermined quantity of MPS signals is a first MPS quantity when the given voltage change is a first voltage change; the predetermined quantity of MPS signals is a second MPS quantity when the given voltage change is a second voltage change; and the first MPS quantity is less than the second MPS quantity and the first voltage change is less than the second voltage change.

12. The method of claim 11, wherein: the predetermined quantity of MPS signals is a third MPS quantity when the given voltage change is a third voltage change; and the third MPS quantity is greater than the second MPS quantity and the third voltage change is greater than the second voltage change.

13. A method, comprising: connecting, through a power-over-Ethernet (POE) interface, to a powered device (PD) over an Ethernet cable; measuring a voltage provided through the Ethernet cable; determining that the voltage has dropped by at least a given voltage change; based on a determination that the voltage has dropped by at least the given voltage change, determining whether or not a predetermined quantity of Maintain Power Signature (MPS) signals have been missed within a given time frame, the predetermined quantity of MPS signals greater than zero; based on a determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determining that the PD is still connected; and based on the determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determining that the PD is still connected, wherein at least one MPS signal was missed within the given time frame.

14. A method, comprising: connecting, through a power-over-Ethernet (POE) interface, to a powered device (PD) over an Ethernet cable; measuring a voltage provided through the Ethernet cable; determining that the voltage has dropped by at least a given voltage change; based on a determination that the voltage has dropped by at least the given voltage change, determining

whether or not a predetermined quantity of Maintain Power Signature (MPS) signals have been missed within a given time frame, the predetermined quantity of MPS signals greater than zero; based on a determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determining that the PD is still connected, wherein the predetermined quantity of MPS signals is based upon the given voltage change; and based on the determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determining that the given voltage change was a transient voltage drop of the apparatus.

15. A method, comprising: connecting, through a power-over-Ethernet (POE) interface, to a powered device (PD) over an Ethernet cable; measuring a voltage provided through the Ethernet cable; determining that the voltage has dropped by at least a given voltage change; based on a determination that the voltage has dropped by at least the given voltage change, determining whether or not a predetermined quantity of Maintain Power Signature (MPS) signals have been missed within a given time frame, the predetermined quantity of MPS signals greater than zero; based on a determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determining that the PD is still connected; based on a determination that at least one MPS signal was missed within the given time frame and that the predetermined quantity of MPS signals has not been missed within the given time frame, determining that the at least one MPS signal was prevented from reaching an apparatus by reverse-biased diodes in the PD.

16. A method, comprising: connecting, through a power-over-Ethernet (POE) interface, to a powered device (PD) over an Ethernet cable; measuring a voltage provided through the Ethernet cable; determining that the voltage has dropped by at least a given voltage change; based on a determination that the voltage has dropped by at least the given voltage change, determining whether or not a predetermined quantity of Maintain Power Signature (MPS) signals have been missed within a given time frame, the predetermined quantity of MPS signals greater than zero; and based on a determination that the predetermined quantity of MPS signals has not been missed within the given time frame, determining that the PD is still connected; wherein: the given time frame is a first time duration when the given voltage change is a first voltage change; the given time frame is a second time duration when the given voltage change is a second voltage change; and the first time duration is less than the second time duration and the first voltage change is less than the second voltage change.

17. An apparatus, comprising: a counter to count missing Maintain Power Signature (MPS) signals; and a control circuit configured to: determine that a voltage provided by the apparatus to a powered device (PD) has dropped by at least a given voltage change; based upon a determination that the voltage provided to the PD has dropped by at least the given voltage change, set the counter to begin counting missing MPS signals; determine whether an additional MPS signal has been received; based upon a determination that the additional MPS signal has been received, determine that the PD is still connected to the apparatus; based upon a determination that the additional MPS signal has not been received, determine whether the counter has reached a predetermined quantity of missing MPS signals; based upon a determination that the counter has reached the predetermined quantity of missing MPS signals, determine that the PD is disconnected from the apparatus.
